

Flexibility and longevity in battery technology

Flexibility and capacity loss

- Design challenges: Flexible battery designs strive to minimise capacity loss caused by flexing, but completely preventing this loss is not possible. In scenarios where batteries are frequently bent, a reduced overall lifetime is expected.

Dependency on use and design

- Battery lifespan: The lifespan of a flexible battery is heavily influenced by its form factor and chemical composition. Batteries with bulkier designs, such as those using solid-state or advanced lithium-ion technologies, generally last longer.
- Application impact: Conversely, thinner batteries typically have shorter lifespans, which can vary depending on their specific uses and the energy demands of these applications.

Energy efficiency in specific applications

- Durability in low-energy uses: In low-power applications, such as RFID tags and sensors, thinner flexible batteries, despite their limitations in flexibility and size, can last several years.

but also align with upcoming PFAS regulations in Europe.

The development process for flexible batteries is complex and typically prolonged, often taking years, and is highly dependent on the specific application—such as e-textiles, which demand batteries that can withstand high temperatures and pressures. Solid-state electrolytes are favoured in these scenarios for their superior temperature resistance. This meticulous process usually involves extensive testing of initial prototypes and the parallel development of future generations, frequently requiring collaboration with other companies to refine designs and meet precise operational needs.

Flexible batteries offer significant adaptability in form, size, and packaging, making them particularly appealing to original equipment manufacturers (OEMs) who require customised battery configurations. The collaboration between battery manufacturers and application companies is vital to develop batteries that meet the diverse requirements of various applications, ensuring both functionality and efficiency.

Moreover, sustainability continues to gain focus. Efforts to enhance the sustainability of battery materials are ongoing, with zinc

increasingly favoured over lithium for its environmental benefits. On the innovation front, significant strides have been made by a French company that developed a glucose-based fuel cell using organic materials. While not yet mainstream, this technology has seen success in the European market, showcasing a shift towards sustainable and recyclable battery technologies. This trend underscores the industry's commitment to addressing environmental issues through innovative advancements, signalling a future where flexible battery technologies not only power systems but do so in an environmentally conscious manner.

Key players in the market

Companies like CCL Design have significantly advanced by leveraging the intellectual property acquired from Imprint Energy, a major zinc-carbon battery producer, to enhance smart labelling and logistics solutions. This strategic acquisition demonstrates how companies can expand their technological capabilities and market presence.

UK-based Zinergy focuses on producing zinc-carbon batteries for similar markets, emphasising the industry's preference for applications that benefit from the batteries'

slim and flexible design. Meanwhile, Blue Spark Technologies highlights its specialisation with the TempTraq product—fever monitoring tags for medical use, showcasing how flexible batteries can address specific health-care needs.

However, many companies in this sector are still developing their technologies. EnSurge, formerly known as Thin Film Power, exemplifies the slow transition from development to market readiness, now moving towards contract securing and product launch.

On the bulkier side, Prologium is shifting from flexible pouch cells to manufacturing battery packs for EVs, reflecting a trend. The new battery packs are made rigid, even though the electrolyte is similar. South Korean companies like Amo Green Tech and LiBest are making headway in the advanced lithium-ion battery market, distinguished by their extensive patents and targeted partnerships.

Across the board, a key strategy that emerges is the focus on working directly with OEMs to develop integrated products rather than merely supplying components. This approach ensures that flexible batteries are tailored to specific applications, enhancing the product's functionality and appeal in targeted markets.

The next big thing

Flexible batteries have found a niche in industries like smart labels and wearables, overcoming early challenges with capacity and form that limited their use around 2010. While not as expansive as sectors like EVs and energy storage systems (ESS), these applications continue to drive flexible battery demand and development.

In the realm of bulk solid-state flexible batteries, some companies initially working with flexible pouch cells have pivoted towards the automotive EV market, utilising the same